

Instance Segmentation

Note: If you are using the factory-installed system image, there is no need to set up the environment from scratch; it is already configured. Simply follow the tutorial to execute the corresponding commands to utilize the features.

1. Instance Segmentation: Images

Use the `yolo11n.pt` model to perform predictions on the sample images included within the Ultralytics project.

Navigate to the code directory:

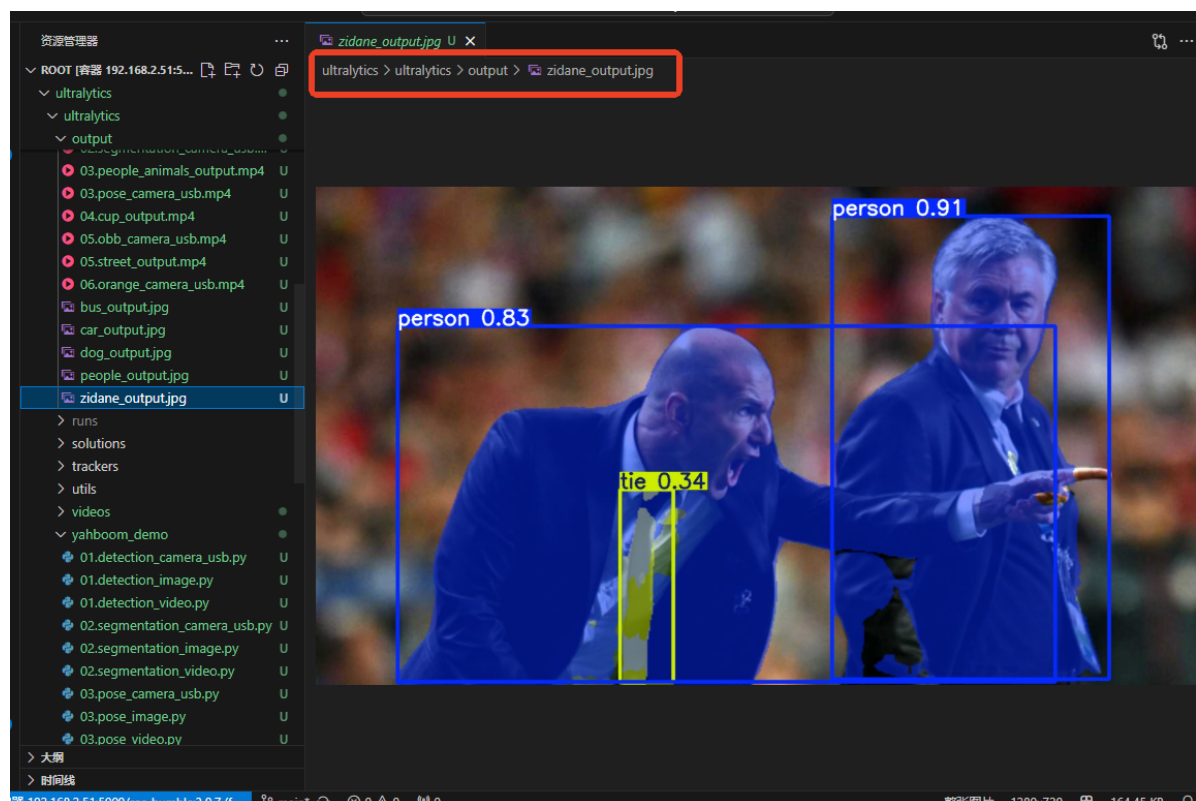
```
cd ~/ultralytics/ultralytics/yahboom_demo
```

Run the code:

```
python3 02.segmentation_image.py
```

Preview of Results

Location of the output images generated by YOLO: `~/ultralytics/ultralytics/output/`



Sample Code:

```
from ultralytics import YOLO

# Load a model
model = YOLO("/home/yahboom/ultralytics/ultralytics/yolo11n-seg.pt") # pretrained yolo11n-seg model
# model = YOLO("/home/yahboom/ultralytics/ultralytics/yolo11n-seg.onnx")
```

```
# model = YOLO("/home/yahboom/ultralytics/ultralytics/yolo11n-seg_ncnn_model")

# Run batched inference on a list of images
results = model("/home/yahboom/ultralytics/ultralytics/assets/zidane.jpg") #
return a list of Results objects

# Process results list
for result in results:
    # boxes = result.boxes # Boxes object for bounding box outputs
    masks = result.masks # Masks object for segmentation masks outputs
    # keypoints = result.keypoints # Keypoints object for pose outputs
    # probs = result.probs # Probs object for classification outputs
    # obb = result.obb # Oriented boxes object for OBB outputs
    result.show() # display to screen

result.save(filename="/home/yahboom/ultralytics/ultralytics/output/zidane_output.jpg") # save to disk
```

2. Instance Segmentation: Video

Use `yolo11n.pt` to perform predictions on a video located within the Ultralytics project directory (specifically, a video *other than* the default examples included with Ultralytics).

Navigate to the code directory:

```
cd /home/yahboom/ultralytics/ultralytics/yahboom_demo
```

Run the code:

```
python3 02.segmentation_video.py
```

Preview of Results

Location of the output video (generated by YOLO detection):

```
/home/yahboom/ultralytics/ultralytics/output/
```



Sample Code:

```
import cv2
from ultralytics import YOLO

# Load the YOLO model
model = YOLO("/home/yahboom/ultralytics/ultralytics/yolo11n-seg.pt")
# model = YOLO("/home/yahboom/ultralytics/ultralytics/yolo11n-seg.onnx")
# model = YOLO("/home/yahboom/ultralytics/ultralytics/yolo11n-seg_ncnn_model")

# Open the video file
video_path = "/home/yahboom/ultralytics/ultralytics/videos/people_animals.mp4"
cap = cv2.VideoCapture(video_path)

# Get the video frame size and frame rate
frame_width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
frame_height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
fps = int(cap.get(cv2.CAP_PROP_FPS))

# Define the codec and create a VideoWriter object to output the processed video
output_path =
"/home/yahboom/ultralytics/ultralytics/output/02.people_animals_output.mp4"
fourcc = cv2.VideoWriter_fourcc(*'mp4v') # You can use 'XVID' or 'mp4v'
depending on your platform
out = cv2.VideoWriter(output_path, fourcc, fps, (frame_width, frame_height))

# Loop through the video frames
while cap.isOpened():
    # Read a frame from the video
    success, frame = cap.read()

    if success:
        # Run YOLO inference on the frame
        results = model(frame)

        # visualize the results on the frame
```

```

annotated_frame = results[0].plot()

# Write the annotated frame to the output video file
out.write(annotated_frame)

# Display the annotated frame
cv2.imshow("YOLO Inference", cv2.resize(annotated_frame, (640, 480)))

# Break the loop if 'q' is pressed
if cv2.waitKey(1) & 0xFF == ord("q"):
    break
else:
    # Break the loop if the end of the video is reached
    break

# Release the video capture and writer objects, and close the display window
cap.release()
out.release()
cv2.destroyAllWindows()

```

3. Instance Segmentation: Real-time Detection

3.1. Launching the Camera

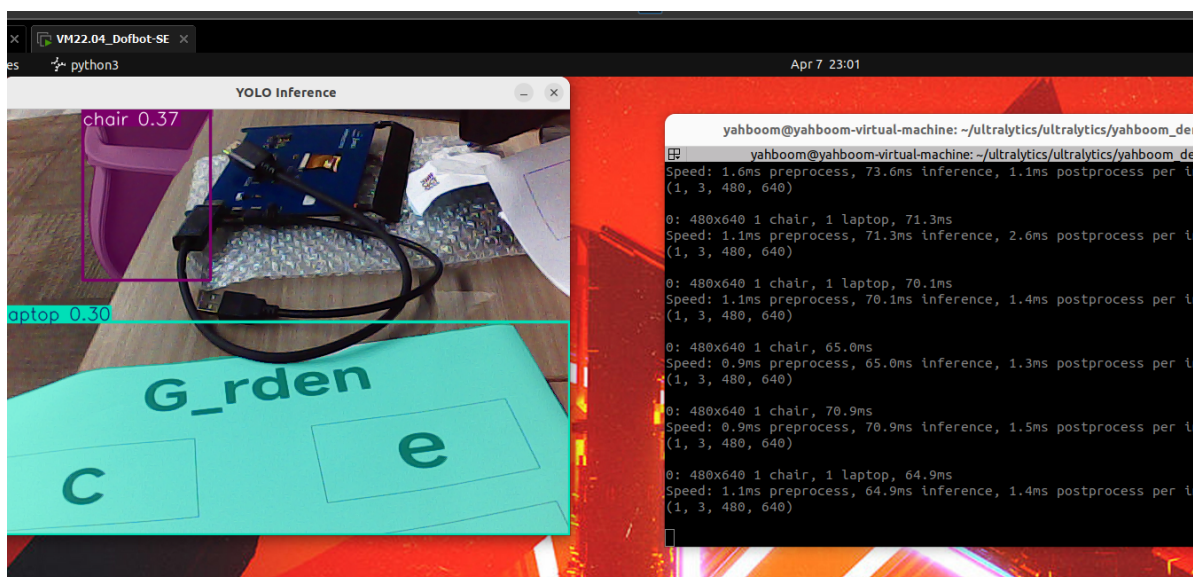
Launch the program below according to your specific camera model. Enter the following command in the terminal:

```
cd /home/yahboom/ultralytics/ultralytics/yahboom_demo
```

Run the code: Click on the preview window, then press the 'q' key to terminate the program!

```
python3 02.segmentation_camera_usb.py
```

Preview



Example Code Path:

/home/yahboom/ultralytics/ultralytics/yahboom_demo/02.segmentation_camera_usb.py

